LAB 4 – Vibe Coding Chat History

Terminal-Based Python 2048 Game

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Purpose

This is a concise record of the LLM/Vibe Coding interaction used to complete Lab 4. It focuses on Tasks 1–4, testing, debugging, and final verification rather than reproducing the entire ChatGPT conversation.

Task 1 – Correct Merge Semantics

User objective

Fix the 2048 merge behavior so each original tile can merge at most once during a move. The key required case was [2,2,2,2] → [4,4,0,0].

LLM implementation

The merge loop was replaced with an index-based implementation that consumes two equal adjacent values at a time.

```
while i < len(values):
  if i + 1 < len(values) and values[i] == values[i + 1]:
    result.append(values[i] * 2)
    i += 2
  else:
    result.append(values[i])
    i += 1
```

Verification

[2,2,2,2] → [4,4,0,0]
[2,2,2,0] → [4,2,0,0]
[2,2,4,4] → [4,8,0,0]
[2,4,4,4] → [2,8,4,0]

Commit: 5578c1e – Fix 2048 merge semantics

Task 2 – Game-State Handling

Inspection showed that the existing implementation already contained the required win detection, no-legal-move detection, and protection against spawning a tile when a move does not change the board. No unnecessary rewrite was made.

Verification

• 2048 state detected correctly.
• Full board with no legal moves detected correctly.
• Full board containing a legal merge remains playable.
• Unchanged move returns False and leaves the board unchanged.

Commit: e3b4e86 – Verify Task 2 game-state handling

Task 3 – Undo and Best Score

Inspection revealed that the original board initialized a score but merge operations were not increasing it. Score accounting was therefore needed for meaningful best-score and undo behavior.

Implementation

The solution saves the board grid and score before a successful move, keeps one undo snapshot, restores both values during undo, and tracks the highest score reached during the current run. Merge operations update the board score.

Tests

• Successful move followed by undo restores board and score.
• Only one level of undo is retained.
• Unchanged moves do not create undo history.
• Best score tracking works correctly.

Commit: a0b3fdb – Add undo and best score

Task 4 – Move Feedback

Requirement

Add concise feedback for successful moves and merges. One accepted move should produce exactly one action-level result, while internal board-scanning functions such as can_move() must not independently print feedback.

Design

The Board API already returns True/False for whether a move changed the board and updates score when merges occur. The implementation therefore calculates the score difference before and after the move.

```
merge_score = self.board.score - old_score
```

Successful moves produce one line such as:
• Moved left.
• Moved left — merged for 4 points.

Invalid and unchanged moves do not claim that a move occurred. Board-level functions remain silent.

Testing

• Successful move with no merge – passed.
• Successful move with merge – passed.
• Unchanged move – passed with no move feedback/state change.
• Invalid command – passed without successful-move feedback.
• Undo regression test – passed.

Commit: 180d961 – Add move feedback

Debugging and Final Fix

After Task 4 was committed, running python3 game.py returned immediately without output. Inspection showed that game.py defined the Game class but did not invoke it when executed directly.

The minimal entry point was added:

```
if __name__ == "__main__":
  Game().run()
```

The project was syntax-checked, the game launched successfully, and the after-gameplay recording was completed. This launcher fix was committed separately so the Task 1–4 commits remained intact.

Final Commit History

Commit	Purpose
Task 4 commit	Add move feedback
a0b3fdb	Add undo and best score
e3b4e86	Verify Task 2 game-state handling
5578c1e	Fix 2048 merge semantics
6a015d8	Update scenario title
38bfa09	Initial commit

Final verification: Tasks 1–4 were tested successfully, the project compiled without syntax errors, the game launched correctly, and the after-gameplay recording was completed.